

Victor J. Huarcaya Azañon

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Summary

Optical engineer currently leading optical AIV for ESA's Comet Interceptor instrument Comet Camera (CoCa), with 10+ years of experience in precision optics, interferometry, and optomechanical systems. Background includes high-stability interferometric testbeds and engineering model development supporting ESA's *LISA* gravitational wave observatory. Skilled in tolerance-driven optical integration, cleanroom workflows (ISO 5/7), and ECSS-compliant verification planning. Strong at the optics–mechanics–electronics interface, with proven ability to translate scientific requirements into traceable, measurable acceptance criteria.

Professional Experience

Optical AIV Engineer

Sept. 2024 – Present

Bern University

Bern, Switzerland

- Lead optical AIV activities for the Comet Camera (CoCa) on ESA's F-class *Comet Interceptor* mission through Phases C/D/E; established and commissioned ISO 5 cleanroom integration facility.
- Delivered flight-representative optical bench on schedule for CDR-level environmental testing, meeting all optical performance requirements (WFE, MTF, co-alignment).
- Authored and released AIT procedures, test scripts, and verification reports under ECSS configuration control; manage non-conformance reports (NCRs) and coordinate rework to closure.
- Drive technical documentation across multi-site AIV teams, ensuring traceability from instrument requirements to acceptance criteria.

Postdoctoral Researcher

Oct. 2023 – Sept. 2024

Max Planck Institute for Gravitational Physics (Albert Einstein Institute)

Hannover, Germany

- *LISA* (Laser Interferometer Space Antenna) Phase Measurement Subsystem and *LISA* engineering model: optical payload Assembly, Integration and Testing.
- Successfully integrated and tested *LISA* Phase Measurement Subsystem, delivering key project milestones for ESA's space-based gravitational wave observatory.

Research Scientist

May 2017 – Oct. 2023

Max Planck Institute for Gravitational Physics (Albert Einstein Institute)

Hannover, Germany

- Achieved world-record sub-picometer structural stability ($< 1 \text{ pm}/\sqrt{\text{Hz}}$ below 1 mHz) in compact interferometers, advancing precision for future space missions.
- Developed 5-DOF test mass optical readout system delivering $10\times$ higher angular precision than commercial autocollimators at low frequencies.
- Designed and validated novel single-element dual-interferometer architecture, reducing system complexity while maintaining sub-picometer performance.
- Operated and maintained ultra-high vacuum systems (10^{-7} mbar) for precision interferometry experiments.

Spacecraft Controller

March 2016 – May 2017

EUMETSAT (for SERCO)

Darmstadt, Germany

- Monitored and commanded *Meteosat-7* spacecraft in 24/7 operational environment, maintaining mission performance through end-of-life phase.
- Executed spacecraft decommissioning procedures as part of mission close-out team.

Research Assistant

Sept. 2013 – Jan. 2016

Centre for Quantum Technologies

Singapore

- Designed, aligned, and characterized electro-optical and optomechanical systems for quantum optics experiments.
- Implemented FPGA-based PID controllers for laser frequency stabilization, improving lock stability and bandwidth.

Education and Certifications

Ph.D. in Physics (Dr. rer. nat.)

Leibniz Universität Hannover

2023

Hannover, Germany

- **Thesis:** *Advancements in Optical Readout Technologies: Test Mass Sensing and Laser-Frequency Stabilization Techniques for Optical Compact Interferometry*
- Research conducted at Max Planck Institute for Gravitational Physics (Albert Einstein Institute)

MSc. in Space Science and Technology

Luleå University of Technology

2013

Luleå, Sweden

- **Thesis:** *Inertial Electrostatic Confinement Devices as an Advanced Electric Propulsion System*
- Research conducted at Institute of Space Systems (IRS) - Stuttgart

MSc. in Space Techniques and Instrumentation

Université Paul Sabatier – Toulouse III

2012

Toulouse, France

Licenciatura in Physics (240 ECTS credits, equivalent to BSc. + MSc.)

University of Balearic Islands

2010

Palma, Spain

ESA's International School on Space Optics – Certification (ESTEC, Noordwijk), 2025

ESA's Space Optics Instrument Design and Technology Course – Certification (Poltu Quatu, Italy), 2023

Laser Safety Officer Certification – LZH Laser Akademie (Hannover, Germany), 2023

Skills

Languages: Spanish (Native) | English (C1 – Advanced) | German (A1 – Basic)

Technical Skills:

- **Opto-Mechanical Integration:**
 - End-to-end AIT of optical payloads: telescope structures, interferometric benches, focal plane assemblies
 - Precision positioning systems: hexapods, piezo stages, goniometers, kinematic mounts
 - Environmental test support: TVAC, thermal cycling, vibration campaigns with pre/post optical verification
 - Tolerance budgeting: flow-down from instrument WFE/MTF requirements to component-level alignment specs
- **Optical Metrology:**
 - Alignment tools: theodolites, autocollimators, shear-plate interferometers, alignment telescopes
 - Wavefront sensing: Shack–Hartmann sensors, Fizeau/Twyman-Green interferometers (Zygo-type), PSF/MTF analysis
 - 3D coordinate metrology: Spatial Analyzer software for theodolite-based measurements
- **Electro-Optics & Detection:**
 - Photodetector systems: single-element PDs, QPDs, low-noise transimpedance amplifiers
 - Laser systems: frequency stabilization, AOM/EOM modulation, Class 3B/4 safety compliance
 - Signal chain: ADC/DAC integration, noise characterization (shot noise, electronic noise floors)
- **Software & Analysis:**
 - Optical design: Zemax OpticStudio (ray tracing, tolerance analysis, optimization)
 - Data analysis & automation: MATLAB, Python (NumPy, SciPy, instrument control APIs)
 - Documentation: LaTeX, MS Office, Adobe Illustrator for technical schematics
 - Lab control: Serial/GPIB/TCP-IP instrument communication, digital/analog I/O
- **Project Management:**
 - Preparation of AIT plans, test procedures, alignment reports, and configuration logs
 - Experience with ECSS / ESA project workflows
 - Procurement and technical coordination with optical and lab-equipment suppliers

Publications

1. **V. Huarcaya** et al. “ 2×10^{-13} fractional laser-frequency stability with a 7-cm unequal-arm Mach-Zehnder interferometer”. *Physical Review Applied*, 20, 024078, 2023. doi:10.1103/PhysRevApplied.20.024078
2. **V. Huarcaya** et al. “Single-Element Dual-Interferometer for Precision Inertial Sensing: Sub-Picometer Structural Stability and Performance as a Reference for Laser Frequency Stabilization”. *Sensors*, 23(24):9758, 2023. doi:10.3390/s23249758
3. **V. Huarcaya** et al. “Five degrees of freedom test mass readout via optical levers”. *Classical and Quantum Gravity*, 37(2):025004, 2019. doi:10.1088/1361-6382/ab5c73
4. Y. Yang, K. Yamamoto, **V. Huarcaya** et al. “Single Element Dual Interferometer for precision inertial sensing”. *Sensors*, 20(17):4986, 2020. doi:10.3390/s20174986
5. S. Aria, [...], **V. Huarcaya** et al. “GLINT – Gravitational-wave laser INterferometry triangle”. *Experimental Astronomy*, 44, 181–208, 2017. doi:10.1007/s10686-017-9558-x
6. **V. Huarcaya**. “Advancements in optical readout technologies: Test mass sensing and laser-frequency stabilization techniques for optical compact interferometry”. *PhD Thesis*, Leibniz Universität Hannover, 2024. doi:10.15488/16938

Selected Presentations

International Conference on Space Optics - ICSO – “Laser-frequency Stabilization via Locking to a Single-Element Mach-Zehnder Interferometer” (Antibes Juan-les-Pins, France), 2024

International Conference on Space Optics - ICSO – “ 2×10^{-13} fractional laser frequency stability with a 7 cm unequal-arm Mach-Zehnder interferometer” (Antibes Juan-les-Pins, France), 2024

13th LISA Symposium – “Five degrees of freedom Test Mass readout via optical levers” (Online), 2020

12th LISA Symposium – “Testing an Optical Readout scheme for future Gravitational Reference Sensors” (Chicago), 2018

Invited Talk – “Demonstration of multi-degree of freedom Test Mass readout” (Wuhan, China), 2018

Alpbach Summer School – “Quantum Physics and Fundamental Physics in Space” (ESA/FFG, Austria), 2015

Scholarships

Graduate Research Scholarship – National University of Singapore, 2014

Erasmus Mundus Mobility Grant – European Commission, 2012

Sa Nostra Foundation Grant – For MSc. studies abroad, Spain, 2010–2011

SENECA Scholarship – Spanish Ministry of Education, 2008

Early Career Highlight

ESA Student Parabolic Flight Campaign – Flight Team Leader for experiment “*Visualization of the Richtmyer-Meshkov instability in zero-g*” (Bordeaux, France), 2004